# 浙江大学2017-18秋冬《数据结构基础》期中模拟练习

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|---|---|---|---|---|---|
| **开始时间** | 1/1/2016, 12:00:00 AM | **结束时间** | 1/18/2038, 12:00:00 AM | **答题时长** | 45分钟 |
| **考生** | – | **得分** | 87 | **总分** | 100 |

## 判断题

总分：15　得分：12

1-1 In a binary search tree, the keys on the same level from left to right must be in sorted (non-decreasing) order. (3分)

◉ T　○ F

评测结果：答案正确（3 分）

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1-2 In a directed graph, the sum of the in-degrees must be equal to the sum of the out-degrees of all the vertices. (3分)

◉ T　○ F

评测结果：答案正确（3 分）

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1-3 If keys are pushed onto a stack in the order {1, 2, 3, 4, 5}, then it is impossible to obtain the output sequence {3, 4, 1, 2, 5}. (3分)

◉ T　○ F

评测结果：答案正确（3 分）

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1-4 For a sequentially stored linear list of length $N$, the time complexities for deleting the first element and inserting the last element are $O(1)$ and $O(N)$, respectively. (3分)

◉ T　○ F

评测结果：答案错误（0 分）

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1-5 $N^2logN$ and $NlogN^2$ have the same speed of growth. (3分)

○ T　◉ F

评测结果：答案正确（3 分）

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## 单选题

总分：65　得分：55

2-1 The recurrent equations for the time complexities of programs P1 and P2 are:

- P1: $T(1)=1, T(N)=T(N/2)+1$;
- P2: $T(1)=1, T(N)=2T(N/2)+1$;

Then the correct conclusion about their time complexities is: (5分)

- (●) A. they are both $O(logN)$
- ( ) B. $O(logN)$ for P1, and $O(N)$ for P2
- ( ) C. they are both $O(N)$
- ( ) D. $O(logN)$ for P1, and $O(NlogN)$ for P2

评测结果：答案错误（0 分）

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2-2 The result of performing three DeleteMin operations in the min-heap {1,3,2,12,6,4,8,15,14,9,7,5,11,13,10} is: (5分)

- ( ) A. 4,5,6,7,8,9,10,11,12,13,14,15
- ( ) B. 4,6,5,13,7,10,8,15,14,12,9,11
- (●) C. 4,6,5,12,7,10,8,15,14,9,13,11
- ( ) D. 4,5,6,12,7,10,8,15,14,13,9,11

评测结果：答案正确（5 分）

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2-3 Given a directed graph G=(V, E) where `V = {v1, v2, v3, v4, v5, v6}` and `E = {<v1,v2>, <v1,v4>, <v2,v6>, <v3,v1>, <v3,v4>, <v4,v5>, <v5,v2>, <v5,v6>}`. Then the topological order of G is: (5分)

- (●) A. `v3, v1, v4, v5, v2, v6`
- ( ) B. `v3, v4, v1, v5, v2, v6`
- ( ) C. `v1, v3, v4, v5, v2, v6`
- ( ) D. `v1, v4, v3, v5, v2, v6`

评测结果：答案正确（5 分）

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2-4 Which of the following statements is TRUE about topological sorting? (5分)

- ( ) A. If a graph has a topological sequence, then its adjacency matrix must be triangular.
- ( ) B. If the adjacency matrix is triangular, then the corresponding directed graph must have a unique topological sequence.
- (●) C. In a DAG, if for any pair of distinct vertices $V_i$ and $V_j$, there is a path either from $V_i$ to $V_j$ or from $V_j$ to $V_i$, then the DAG must have a unique topological sequence.
- ( ) D. If $V_i$ precedes $V_j$ in a topological sequence, then there must be a path from $V_i$ to $V_j$.

评测结果：答案正确（5 分）

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2-5 To delete `p` from a doubly linked list, we must do: (6分)

- A. `p->prior=p->prior->prior; p->prior->next=p;`
- B. `p->next->prior=p; p->next=p->next->next;`
- C. `p->prior->next=p->next; p->next->prior=p->prior;`
- D. `p->next=p->prior->prior; p->prior=p->next->next;`

评测结果：答案正确（6 分）

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2-6 In-order traversal of a binary tree can be done iteratively. Given the stack operation sequence as the following:

```
push(1), push(2), push(3), pop(), push(4), pop(), pop(), push(5), pop(), pop(), push(6), pop(
```

Which one of the following statements is TRUE? (6分)

- A. 6 is the root
- B. 2 is the parent of 4
- C. 2 and 6 are siblings
- D. None of the above

评测结果：答案正确（6 分）

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2-7 Given a quadtree(四叉树) with 2 nodes of degree 2, 3 nodes of degree 3, 4 nodes of degree 4. The number of leaf nodes in this tree is __. (5分)

- A. 10
- B. 12
- C. 20
- D. 21

评测结果：答案错误（0 分）

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2-8 The array representation of a disjoint set is given by { 4, 6, 5, 2, -3, -4, 3 }. If the elements are numbered from 1 to 7, the resulting array after invoking `Union(Find(7),Find(1))` with union-by-size and path-compression is: (5分)

- A. { 4, 6, 5, 2, 6, -7, 3 }
- B. { 4, 6, 5, 2, -7, 5, 3 }
- C. { 6, 6, 5, 6, -7, 5, 5 }
- D. { 6, 6, 5, 6, 6, -7, 5 }

评测结果：答案正确（5 分）

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2-9 If an undirected graph G = (V, E) contains 7 vertices. Then to guarantee that G is connected in any cases, there has to be at least __ edges. (6分)

- A. 6
- B. 15
- C. 16
- D. 21

评测结果：答案正确（6 分）

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2-10 Insert { 5, 11, 13, 1, 3, 6 } one by one into an initially empty binary search tree. The post-order traversal sequence of the resulting tree is: (6分)

- A. 3, 1, 5, 6, 13, 11
- B. 3, 1, 6, 13, 11, 5
- C. 1, 3, 11, 6, 13, 5
- D. 1, 3, 5, 6, 13, 11

评测结果：答案正确（6 分）

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2-11 Suppose that an array of size `m` is used to store a circular queue. If the head pointer `front` and the current size variable `size` are used to represent the range of the queue instead of `front` and `rear`, then the maximum capacity of this queue can be: (5分)

- A. `m-1`
- B. `m`
- C. `m+1`
- D. cannot be determined

评测结果：答案正确（5 分）

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2-12 How many leaf node does a complete binary tree with 2435 nodes have? (6分)

- A. 1218
- B. 1217
- C. 812
- D. cannot be determined

评测结果：答案正确（6 分）

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## 程序填空题

总分：20　得分：20

5-1 The function is to find the `K`-th smallest element in a list `A` of `N` elements. The function `BuildMaxHeap(H, K)` is to arrange elements `H[1]` ... `H[K]` into a max-heap. Please complete the following program.

```
ElementType FindKthSmallest ( int A[], int N, int K )
{   /* it is assumed that K<=N */
    ElementType *H;
    int i, next, child;

    H = (ElementType *)malloc((K+1)*sizeof(ElementType));
    for ( i=1; i<=K; i++ ) H[i] = A[i-1];
    BuildMaxHeap(H, K);

    for ( next=K; next<N; next++ ) {
        H[0] = A[next];
        if ( H[0] < H[1] ) {
            for ( i=1; i*2<=K; i=child ) {
                child = i*2;
                if ( child!=K && H[child+1]>H[child]          (5分) ) child++;
                if ( H[child]>H[0]                            (5分) )
                    H[i] = H[child];
                else break;
            }
            H[i] = H[0];
        }
    }
    return H[1];
}
```

评测结果：答案正确（10 分）

| 序号 | 结果 | 得分 |
|---|---|---|
| 0 | 答案正确 | 5 |
| 1 | 答案正确 | 5 |

5-2 Please fill in the blanks in the program which performs `Find` as a Union/Find operation with path compression.

```
SetType Find ( ElementType X, DisjSet S )
{
   ElementType root, trail, lead;
```

```
    for ( root = X; S[root] > 0; root=S[root]          (5分) ) ;
    for ( trail = X; trail != root; trail = lead ) {
       lead = S[trail] ;
       S[trail]=root                (5分);
    }
    return root;
}
```

评测结果：答案正确（10 分）

| 序号 | 结果 | 得分 |
| --- | --- | --- |
| 0 | 答案正确 | 5 |
| 1 | 答案正确 | 5 |